Review Questions

1. Advantages to creating methods in addition to the `main()` method include all of the following *except* _____.

    a. the `main()` method is easier to follow

    b. all methods are prewritten, saving the programmer work

    c. it is easier to determine the overall intent of statements encapsulated in the methods

    d. methods are reusable

    Answer: b

    Feedback:

    All methods are not prewritten, so that is not an advantage to creating them. Making the `main()` method easier to follow, making it easier to determine the overall intent of multiple code lines, and being able to reuse methods are all advantages to creating them.

2. In Java, methods must include all of the following *except* _____.

    a. a call to another method

    b. a declaration

    c. curly braces

    d. a body

    Answer: a

    Feedback:

    Java methods do not need to include calls to other methods. However, Java methods must have a declaration and must have curly braces that surround the method body, although the body can be empty.

3. All method declarations contain _____.

    a. arguments

    b. one or more explicitly named access specifiers

c. parentheses

d. the keyword static

Answer: c

Feedback:

All method declarations contain parentheses following the method name. Not all method declarations contain arguments, although some do. Not all methods contain explicitly named access specifiers, although some do. Not all method declarations contain the keyword `static`, although some do.

4. A public static method named `computeSum()` is located in `ClassA`. To call the method from within `ClassB`, use the statement _____.

   a. `ClassA.computeSum();`

   b. `ClassB(computeSum());`

   c. `ComputeSum(ClassA);`

   d. You cannot call `computeSum()` from within `ClassB`.

   Answer: a

   Feedback:

   You use the name of the class that holds the method and the method name: `ClassA.computeSum();`. The other two statements will not compile.

5. Which of the following method declarations is correct for a `static` method named `displayFacts()` if the method receives an `int` argument?

   a. `public static int displayFacts()`

   b. `public void displayFacts(int data)`

   c. `public static void displayFacts(int data)`

   d. `public void displayFacts(static int)`

   Answer: c

   Feedback:

The method is `static` and receives an `int` argument. The other answers are incorrect because one does not receive an `int` argument, one is not `static`, and an argument cannot be `static` and it must have an identifier.

6. Which of the following method declarations is correct for a `static` method named `computeSum()` if the method receives two `double` arguments?

    a. `public static double, double computeSum()`

    b. `public static void computeSum(double a, double b)`

    c. `public int computeSum(double a, double b)`

    d. `public static int computeSum(double a, b)`

    Answer: b

    Feedback:

    The static method `computeSum()` receives two `double` arguments. The other answers are wrong because a method cannot have two return types, the method must use the keyword `static`, and a data type must be specified for each argument, even if each is the same data type.

7. The method with the declaration `public static int aMethod(double d)` is a method type of _____.

    a. `static`

    b. `int`

    c. `double`

    d. You cannot determine the method type.

    Answer: b

    Feedback:

    The method's type is `int`. You can determine the method type, but `static` is not a type. The method's type is not `double`, but it does take a `double` argument.

8. Which of the following is a correct call to a method declared as `public static void aMethod(char code)`?

    a. `void aMethod();`

b. `void aMethod('V');`

c. `aMethod(char 'M');`

d. `aMethod('Q');`

Answer: d

Feedback:

You call the method using its name and a `char` argument within parentheses. You do not precede a method call with `void`, and the method call requires an argument.

9. A method is declared as `public static void showResults(double d, int i)`. Which of the following is a correct method call?

a. `showResults(double d, int i);`

b. `showResults(12.2, 67);`

c. `showResults(4, 99.7);`

d. Two of these are correct.

Answer: b

Feedback:

You can call the method using 12.2 and 67 as arguments. You do not use data types in a method call. The argument 4 can be passed to the `double`, but 99.7 cannot be passed to the `int`.

10. The method with the declaration `public static char procedure(double d)` has a method type of _____.

a. `public`

b. `static`

c. `char`

d. `double`

Answer: c

Feedback:

The method's data type is `char`. A method's data type is the type of value it returns. Neither `public` nor `static` is a data type. The method's parameter type is `double`, but the method's type is not `double`.

11. The method `public static boolean testValue(int response)` returns _____.

    a. no value

    b. an `int` value

    c. a `boolean` value

    d. You cannot determine what is returned.

    Answer: c

    Feedback:

    The method returns a `boolean` value. The method takes an `int` argument, but it does not return an `int`.

12. Which of the following could be the last legally coded line of a method declared as `public static int getVal(double sum)`?

    a. `return;`

    b. `return 77;`

    c. `return 2.3;`

    d. Any of these could be the last coded line of the method.

    Answer: b

    Feedback:

    The method returns an integer, and 77 is an integer. The return statement that returns nothing does not work, and neither does the one that returns a `double`.

13. In the method header `public static boolean(int age)`, `age` is a(n) ____.

    a. argument

    b. parameter

    c. `return` value

d. `final` value

Answer: b

Feedback:

The variable `age` is a parameter. An argument appears in a method call, not in a method header. The variable `age` is neither a `return` value nor a `final` value.

14. The code between a pair of curly braces in a method is a _____.

a. function

b. brick

c. block

d. sector

Answer: c

Feedback:

A block exists between curly braces. A function is similar to a method in some programming languages. Neither “brick” nor “sector” is a term used in Java programming.

15. When a block exists within another block, the blocks are _____.

a. structured

b. illegal

c. sheltered

d. nested

Answer: d

Feedback:

Blocks are nested when one exists inside the other; it is legal. “Structured” is a term that applies to programming logic. “Sheltered” is not a term used in Java programming.

16. The portion of a program within which you can reference a variable is the variable’s _____.

a. scope

b. space

c. domain

d. range

Answer: a

Feedback:

A variable can be referenced within its scope. Although “domain,” “range,” and “space” all have approximately the same meaning in English, they are not the words programmers use to describe the portion of a program in which you can refer to a variable.

17. You can declare variables with the same name multiple times _____.

a. within a statement

b. within a block

c. within a method

d. You never can declare multiple variables with the same name.

Answer: c

Feedback:

You can declare variables with the same name within a method if the declarations are in different blocks You cannot declare variables with the same name within a statement or block.

18. Nonambiguous, overloaded methods must have the same _____.

a. types of parameters

b. number of parameters

c. parameter names

d. name

Answer: d

Feedback:

By definition, nonambiguous overloaded methods have the same name. Nonambiguous overloaded methods can have the same types or number of parameters, but the lists must be different. Nonambiguous overloaded methods can have the same parameter names, but the parameter names can be different.

19. If a method is written to receive a `double` parameter, and you pass an `int` to the method, then the method will _____.

    a. work correctly; the `int` will be promoted to a `double`

    b. work correctly; the parameter type will automatically become an `int`

    c. execute but issue a warning

    d. not work; an error message will be issued

    Answer: a

    Feedback:

    The `int` will be promoted to a `double`. The method will work, but the parameter will not become an `int`; it will become a `double`. No warning or error message is issued.

20. If a method is written to receive an `int` parameter, and you pass a `double` to the method, then the method will _____.

    a. work correctly; the `double` will be promoted to an `int`

    b. work correctly, but the `double` might lose some data

    c. execute but issue a warning

    d. not work; an error message will be issued

    Answer: d

    Feedback:

    The method will not work.; an error message will be displayed.